IT Employee Lifecycle Automation System

Functional Requirements Specification

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1. Stakeholders

Stakeholder	Interest in the System
IT Administrators	Primary users who manage account lifecycle operations, configure automation rules, handle exceptions, and ensure security compliance. They benefit from reduced manual workload and fewer errors.
HR Personnel	Staff who maintain employee records in OrangeHRM and need visibility into IT provisioning status to confirm that new hires receive timely access and departing employees are offboarded.
Department Managers	Leaders responsible for ensuring their new hires have appropriate access on day one and that departing team members are properly offboarded. They may request additional permissions for team members.
Compliance Officers	Reviewers who require audit trails to verify that account access changes follow organizational policies and regulatory requirements.
CIO / IT Director	Project sponsor who seeks reduced operational cost, improved security posture, and elimination of orphaned accounts through automation.
Employees (End Users)	Indirect beneficiaries who receive timely account access upon hiring, ensuring productive first days without waiting for manual IT provisioning.

2. Actors and Goals

Primary Actors

- **IT Administrator:** This actor configures automation rules, manages department-to-security-group mappings, approves or overrides automated account actions, views audit logs, generates compliance reports, and searches employee account status. The IT Administrator has full control over system settings and can intervene in any automated workflow.

- **HR Personnel:** This actor views the provisioning status of newly entered employees, flags high-priority onboarding requests for expedited processing, confirms successful completion of offboarding procedures, and accesses a read-only view of employee account status.
- **Department Manager:** This actor views account status for their direct reports, submits requests for additional access or permissions for team members, and receives notifications when team member accounts are provisioned or modified.

Secondary Actors

- **System (Automation Engine):** This actor is responsible for polling the OrangeHRM REST API at configured intervals to detect new hires and terminations, automatically creating or disabling Active Directory accounts, assigning security groups based on configured role-permission mappings, sending notifications to stakeholders, and retrying failed operations.
- **OrangeHRM (External System):** The human resource management system that serves as the authoritative source of employee data. The automation engine queries its REST API to detect employee status changes.
- **Active Directory (External System):** The directory service that receives account creation, modification, and disabling commands from the automation engine via LDAP protocol.

3. Use Cases

3.1 Casual Descriptions

IT Administrator (total: 25 points)

Use Case	Description	Est.
Login/Logout	Authenticate to the system using credentials and manage session with role-based access control	4
View Dashboard	View a real-time dashboard displaying pending, in-progress, completed, and failed onboarding/offboarding tasks with statistics	2
Configure Role-Permission Mappings	Set up and maintain mappings between HR departments/job roles and Active Directory security groups to drive automated group assignments	4
Approve/Override Account Actions	Review pending automated account actions and manually approve, modify, or reject them before execution	4
Configure Polling Interval	Set the frequency at which the system polls OrangeHRM for	2

	employee status changes	
View Audit Logs	Browse and search a chronological record of all account lifecycle events for compliance and troubleshooting	3
Generate Compliance Reports	Create and export compliance or activity summary reports for specified date ranges and department filters	4
Search Employee Status	Look up any employee by name, ID, or email and view their current Active Directory account status	2

HR Personnel (total: 8 points)

Use Case	Description	Est.
View Provisioning Status	Check the current provisioning status of newly entered employees to confirm account creation is underway	2
Flag High-Priority Onboarding	Mark an onboarding request as urgent so the system processes it ahead of normal-priority tasks	2
Confirm Offboarding	Verify and acknowledge that offboarding procedures have been successfully completed for a departed employee	2
View Employee Account Status	Access a read-only view of any employee's current Active Directory account status	2

Department Manager (total: 7 points)

Use Case	Description	Est.
View Team Account Status	View the current account provisioning status for all direct reports in the department	2
Request Additional Access	Submit a request for additional security groups or permissions for a team member, routed to IT for approval	3

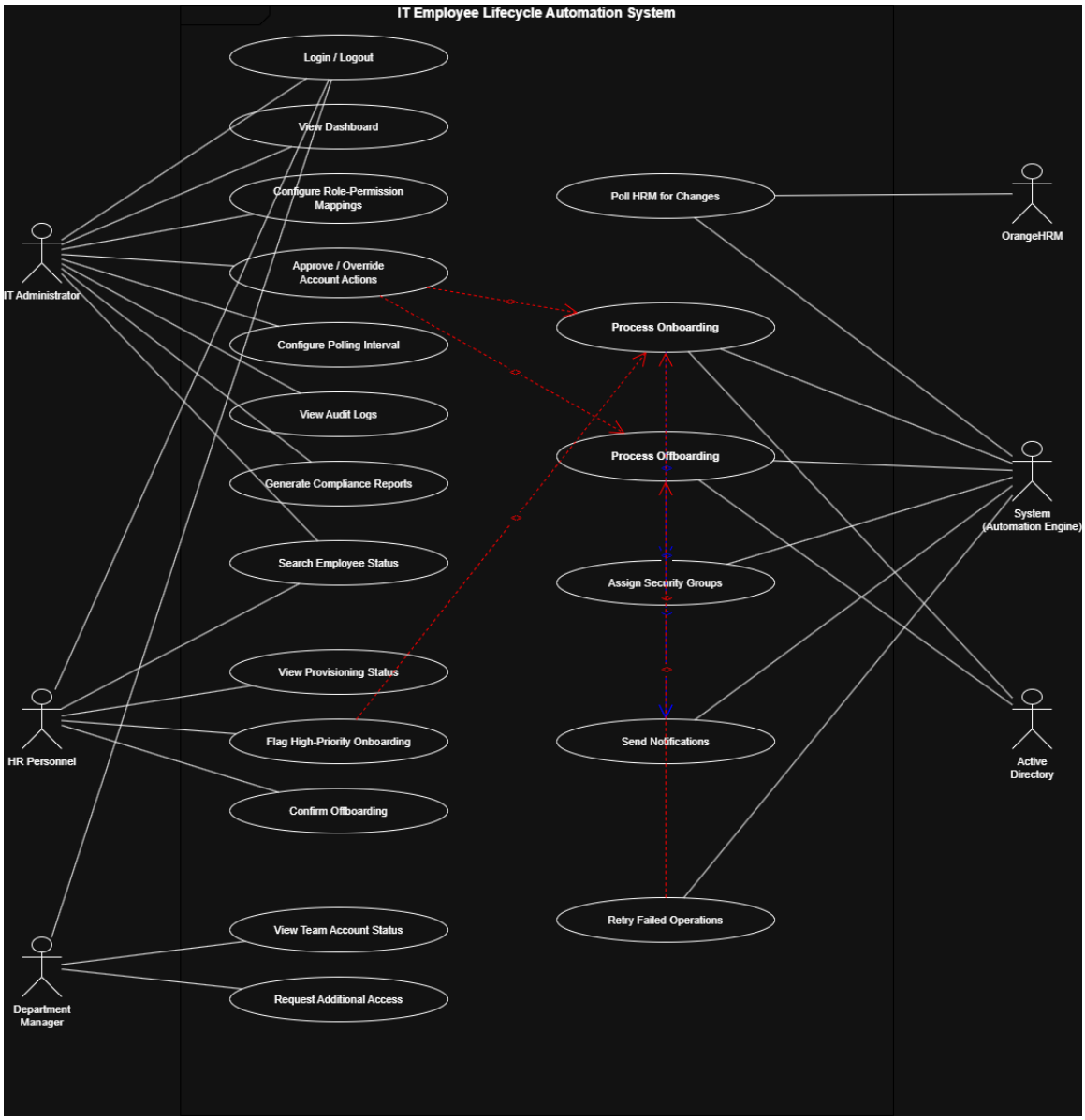
Receive Notifications	Receive alerts when team member accounts are provisioned, modified, or when actions require attention	2

System — Automation Engine (total: 18 points)

Use Case	Description	Est.
Poll HRM for Changes	Periodically query the OrangeHRM REST API to detect new hires and terminations based on employee status changes	3
Process Onboarding	Create a new Active Directory account for a detected new hire and trigger security group assignment and notifications	4
Process Offboarding	Disable the Active Directory account and revoke group memberships for a detected terminated employee	3
Assign Security Groups	Automatically assign AD security groups to a new account based on the employee's department and role mappings	3
Send Notifications	Dispatch email or in-app notifications to relevant stakeholders when account actions complete or fail	3
Retry Failed Operations	Automatically retry a failed account operation up to 3 times before marking the task as permanently failed	2

Grand Total: 58 points (~29 engineer-days

3.2 Use Case Diagram

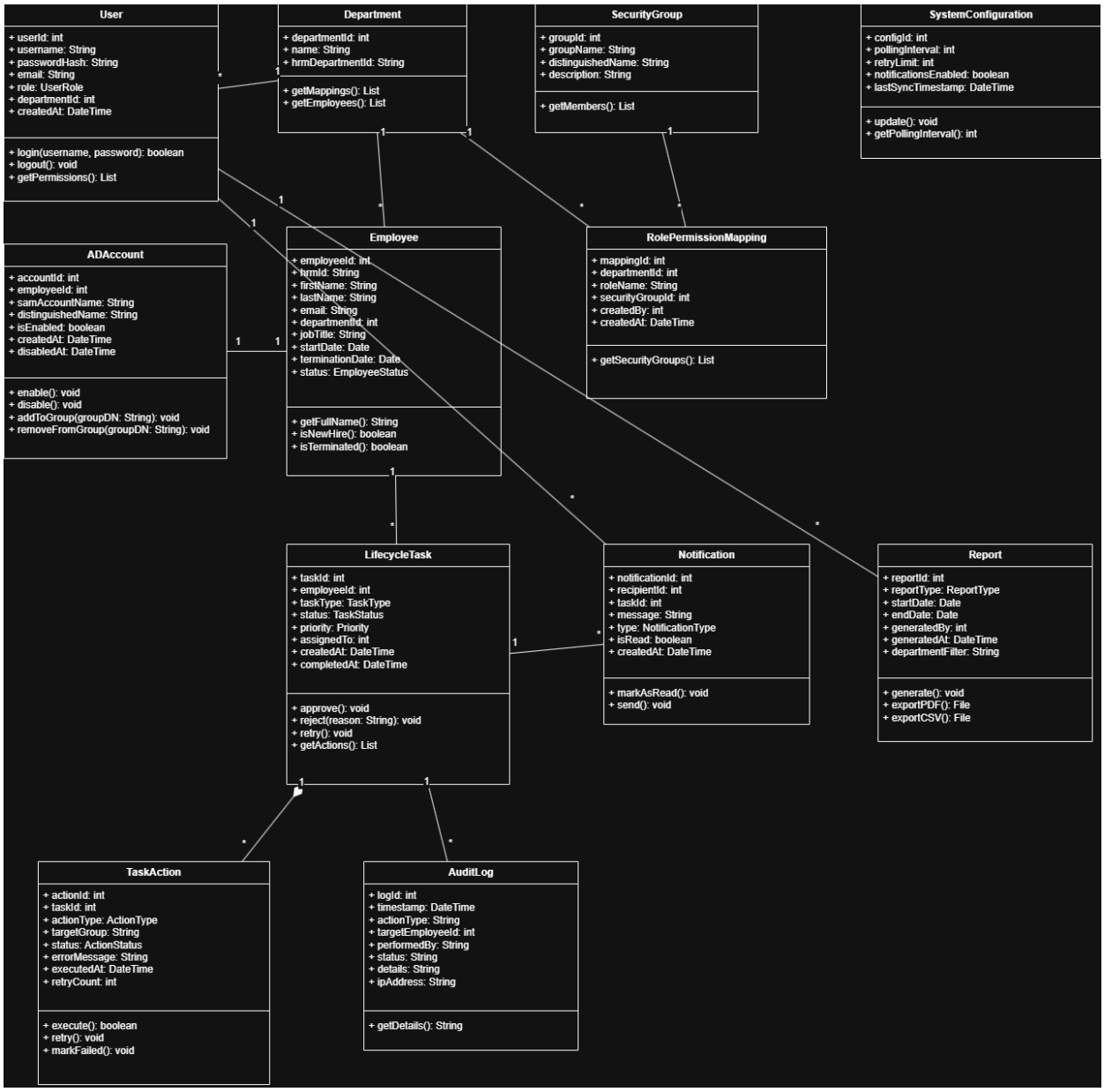


- Process Onboarding **<<include>>** Assign Security Groups — Every onboarding task always assigns security groups based on department/role mappings.
- Process Onboarding **<<include>>** Send Notifications — Stakeholders are always notified after an onboarding action completes or fails.
- Process Offboarding **<<include>>** Send Notifications — Stakeholders are always notified after an offboarding action completes or fails.

<<extend>> relationships (optional/conditional behavior):

- Approve/Override Account Actions **<<extend>>** Process Onboarding — An IT administrator may optionally intervene to approve or modify actions before onboarding executes.
- Approve/Override Account Actions **<<extend>>** Process Offboarding — An IT administrator may optionally intervene before offboarding executes.
- Flag High-Priority Onboarding **<<extend>>** Process Onboarding — HR personnel may optionally flag a task for expedited processing.
- Retry Failed Operations **<<extend>>** Process Onboarding — The system retries only if the onboarding operation fails.
- Retry Failed Operations **<<extend>>** Process Offboarding — The system retries only if the offboarding operation fails.

4. Class Diagram



4.1 Component Overview

Using a bottom-up approach, we first identify the smaller components of the system (employees, accounts, security groups), then build up to the full lifecycle automation system.

Category	Classes
Core Entities	Employee, ADAccount, Department, SecurityGroup
Configuration	RolePermissionMapping, SystemConfiguration
Workflow	LifecycleTask, TaskAction
Audit and Reporting	AuditLog, Notification, Report
Access Control	User

4.2 Enumerations

Enumeration	Values	Description
UserRole	IT_ADMIN, HR_PERSONNEL, DEPT_MANAGER,	Defines the role-based access level of a system user

	COMPLIANCE_OFFICER	
EmployeeStatus	ACTIVE, TERMINATED, PENDING	Represents the current lifecycle state of an employee
TaskType	ONBOARD, OFFBOARD	Categorizes a lifecycle task as either onboarding or offboarding
TaskStatus	PENDING, IN_PROGRESS, COMPLETED, FAILED	Tracks the execution state of a lifecycle task
ActionType	CREATE_ACCOUNT, DISABLE_ACCOUNT, ASSIGN_GROUP, REMOVE_GROUP	Specifies the type of Active Directory operation
ActionStatus	PENDING, EXECUTING, SUCCESS, FAILED	Tracks the execution state of an individual task action
Priority	NORMAL, HIGH	Indicates the processing priority of a lifecycle task
NotificationType	INFO, WARNING, ERROR	Categorizes the severity of a notification
ReportType	COMPLIANCE, ACTIVITY_SUMMARY, FAILED_ACTIONS	Specifies the type of report to generate

4.3 Data Types and Operation Signatures

Class: User

A system user who accesses the application. Each user has a role that determines their permissions and available features.

Attribute	Type	Description
userId	int	Unique identifier for the user
username	String	Login username for authentication
passwordHash	String	Bcrypt-hashed password
email	String	Email address for notifications
role	UserRole	The user's role determining access level
departmentId	int	Foreign key referencing the user's department
createdAt	DateTime	Timestamp when the user account was created

Operation	Signature	Description
login	login(username: String, password: String): boolean	Authenticates the user credentials and creates a session; returns true on success
logout	logout(): void	Terminates the user's active session
getPermissions	getPermissions(): List	Returns the list of permission strings associated with the user's role

Class: Employee

An employee record synchronized from OrangeHRM. Represents a person in the organization whose account lifecycle is managed by this system.

Attribute	Type	Description
employeeId	int	Unique identifier in this system
hrmId	String	The employee's unique identifier in OrangeHRM
firstName	String	Employee's first name
lastName	String	Employee's last name
email	String	Employee's work email address
departmentId	int	Foreign key referencing the employee's department
jobTitle	String	The employee's job title/role
startDate	Date	The employee's hire date
terminationDate	Date	The employee's termination date (null if active)
status	EmployeeStatus	Current lifecycle state: ACTIVE, TERMINATED, or PENDING

Operation	Signature	Description
getFullName	getFullName(): String	Returns the concatenation of firstName and lastName
isNewHire	isNewHire(): boolean	Returns true if the employee was added within the current polling cycle
isTerminated	isTerminated(): boolean	Returns true if the employee's status is TERMINATED

Class: Department

An organizational department imported from OrangeHRM. Departments are used as a key dimension for role-permission mappings.

Attribute	Type	Description
departmentId	int	Unique identifier for the department
name	String	Display name of the department
hrmDepartmentId	String	The department's unique identifier in OrangeHRM

Operation	Signature	Description
getMappings	getMappings(): List	Returns all role-permission mappings associated with this department
getEmployees	getEmployees(): List	Returns all employees belonging to this department

Class: SecurityGroup

An Active Directory security group that controls access to resources. Groups are assigned to employee accounts based on role-permission mappings.

Attribute	Type	Description
groupId	int	Unique identifier for the security group
groupName	String	Display name of the AD security group
distinguishedName	String	The full LDAP distinguished name (DN) of the group in Active Directory
description	String	A brief description of the group's purpose

Operation	Signature	Description
getMembers	getMembers(): List	Returns all AD accounts currently assigned to this security group

Class: RolePermissionMapping

A configuration entry that maps a department and job role combination to a specific Active Directory security group. These mappings drive the automatic group assignment during onboarding.

Attribute	Type	Description
mappingId	int	Unique identifier for the mapping
departmentId	int	Foreign key referencing the department
roleName	String	The job title/role this mapping applies to
securityGroupId	int	Foreign key referencing the target security group
createdBy	int	Foreign key referencing the user who created this mapping
createdAt	DateTime	Timestamp when the mapping was created

Operation	Signature	Description
getSecurityGroups	getSecurityGroups(): List	Returns all security groups mapped to this department-role combination

Class: LifecycleTask

A workflow task representing an onboarding or offboarding operation for a specific employee. Tasks progress through statuses and contain one or more individual actions.

Attribute	Type	Description
taskId	int	Unique identifier for the task
employeeId	int	Foreign key referencing the target employee
taskType	TaskType	ONBOARD or OFFBOARD
status	TaskStatus	Current execution state: PENDING, IN_PROGRESS, COMPLETED, or FAILED
priority	Priority	Processing priority: NORMAL or HIGH
assignedTo	int	Foreign key referencing the user assigned to review (null if automated)

createdAt	DateTime	Timestamp when the task was created
completedAt	DateTime	Timestamp when the task was completed (null if not yet completed)

Operation	Signature	Description
approve	approve(): void	Marks the task as approved and triggers execution of all pending actions
reject	reject(reason: String): void	Cancels the task with a documented reason
retry	retry(): void	Resets failed actions and re-executes the task
getActions	getActions(): List	Returns all individual actions belonging to this task

Class: TaskAction

An individual atomic action within a lifecycle task, such as creating an account or assigning a security group. Each action tracks its own execution status and retry count.

Attribute	Type	Description
actionId	int	Unique identifier for the action
taskId	int	Foreign key referencing the parent lifecycle task
actionType	ActionType	The type of AD operation: CREATE_ACCOUNT, DISABLE_ACCOUNT, ASSIGN_GROUP, or REMOVE_GROUP
targetGroup	String	The DN of the target security group (null for account create/disable actions)
status	ActionStatus	Execution state: PENDING, EXECUTING, SUCCESS, or FAILED
errorMessage	String	Error details if the action failed (null on success)
executedAt	DateTime	Timestamp when the action was last executed
retryCount	int	Number of retry attempts made (max 3)

Operation	Signature	Description
execute	execute(): boolean	Performs the AD operation; returns true on success
retry	retry(): void	Increments retryCount and re-executes the action
markFailed	markFailed(): void	Sets status to FAILED after exhausting retry attempts

Class: AuditLog

A chronological record of an account lifecycle event. Audit logs provide an immutable trail for compliance and troubleshooting.

Attribute	Type	Description
logId	int	Unique identifier for the log entry
timestamp	DateTime	When the event occurred
actionType	String	Description of the action performed
targetEmployeeId	int	Foreign key referencing the affected employee
performedBy	String	Username of the actor or "System" for automated actions
status	String	Outcome of the action (e.g., SUCCESS, FAILED)
details	String	Additional context or error messages
ipAddress	String	IP address from which the action was initiated

Operation	Signature	Description
getDetails	getDetails(): String	Returns the full detail string for this log entry

Class: Notification

An alert sent to a system user when account actions are completed, failed, or require attention.

Attribute	Type	Description
notificationId	int	Unique identifier for the

		notification
recipientId	int	Foreign key referencing the recipient user
taskId	int	Foreign key referencing the related lifecycle task
message	String	The notification message body
type	NotificationType	Severity level: INFO, WARNING, or ERROR
isRead	boolean	Whether the recipient has read the notification
createdAt	DateTime	Timestamp when the notification was created

Operation	Signature	Description
markAsRead	markAsRead(): void	Sets isRead to true
send	send(): void	Dispatches the notification to the recipient via email or in-app alert

Class: Report

A generated compliance or activity report covering a specified date range and optional department filter.

Attribute	Type	Description
reportId	int	Unique identifier for the report
reportType	ReportType	The type of report: COMPLIANCE, ACTIVITY_SUMMARY, or FAILED_ACTIONS
startDate	Date	Beginning of the report date range
endDate	Date	End of the report date range
generatedBy	int	Foreign key referencing the user who generated the report
generatedAt	DateTime	Timestamp when the report was generated
departmentFilter	String	Comma-separated department IDs to filter by (null for all departments)

Operation	Signature	Description
generate	generate(): void	Queries audit logs and task data to populate the report content
exportPDF	exportPDF(): File	Renders the report as a downloadable PDF document
exportCSV	exportCSV(): File	Exports the report data as a CSV file

Class: SystemConfiguration

Global system settings that control automation behavior, including polling frequency and retry limits.

Attribute	Type	Description
configId	int	Unique identifier for the configuration record
pollingInterval	int	Interval in minutes between HRM synchronization polls
retryLimit	int	Maximum number of retry attempts for failed operations
notificationsEnabled	boolean	Whether the system sends notifications on task completion/failure
lastSyncTimestamp	DateTime	Timestamp of the most recent successful HRM synchronization

Operation	Signature	Description
update	update(): void	Persists changes to the configuration settings
getPollingInterval	getPollingInterval(): int	Returns the current polling interval in minutes

Class: ADAccount

An Active Directory user account managed by this system. Each ADAccount is linked to exactly one Employee.

Attribute	Type	Description
accountId	int	Unique identifier for the AD account record
employeeId	int	Foreign key referencing the associated employee

samAccountName	String	The AD sAMAccountName (login name)
distinguishedName	String	The full LDAP distinguished name of the account
isEnabled	boolean	Whether the account is currently enabled in Active Directory
createdAt	DateTime	Timestamp when the AD account was created
disabledAt	DateTime	Timestamp when the account was disabled (null if active)

Operation	Signature	Description
enable	enable(): void	Enables the AD account via LDAP
disable	disable(): void	Disables the AD account via LDAP and records the timestamp
addToGroup	addToGroup(groupDN: String): void	Adds this account to the specified AD security group
removeFromGroup	removeFromGroup(groupDN: String): void	Removes this account from the specified AD security group

4.4 Class Relationships

Relationship	Type	Multiplicity	Description
User — Department	Association	Many-to-one (*..1)	Each User belongs to one Department; a Department may have many Users
Employee — Department	Association	Many-to-one (*..1)	Each Employee belongs to one Department; a Department may have many Employees
Employee — ADAccount	Association	One-to-one (1..1)	Each Employee has at most one ADAccount; each ADAccount belongs to exactly one Employee
Employee — LifecycleTask	Association	One-to-many (1..*)	Each Employee may have multiple LifecycleTasks over

			time; each task references one Employee
Department — RolePermissionMapping	Association	One-to-many (1..*)	Each Department may have multiple mappings for different roles; each mapping belongs to one Department
RolePermissionMapping — SecurityGroup	Association	Many-to-one (*..1)	Each mapping references one SecurityGroup; a SecurityGroup may be referenced by many mappings
LifecycleTask — TaskAction	Composition	One-to-many (1..*)	Each LifecycleTask contains one or more TaskActions; actions cannot exist without their parent task
LifecycleTask — AuditLog	Association	One-to-many (1..*)	Each LifecycleTask generates one or more AuditLog entries
LifecycleTask — Notification	Association	One-to-many (1..*)	Each LifecycleTask may trigger one or more Notifications
User — Notification	Association	One-to-many (1..*)	Each User may receive multiple Notifications
User — Report	Association	One-to-many (1..*)	Each User may generate multiple Reports